How different might things had been had Bayesian statistical science been available to the most famous scientists in history

by S. James Press
and Judith M. Tanur [1]

It might be of interest to conjecture a bit about what 12 of the famous scientists in history, from Aristotle to Einstein, might have done had Bayesian analysis been available to them. (A detailed rationale for how these 12 scientists were selected is provided in Press and Tanur, 2001.) Suppose that advanced mathematics, subjective probability, and Bayesian statistical methods of analyzing scientific data had already been developed and were readily available at the time some of these scientists lived and worked. How might that research methodology have been employed to strengthen the positions that those scientists held, thus perhaps settling more definitively some of the questions still raised, even today, perhaps hundreds of years later? In many cases we are still questioning their research methodologies, their judgments, their conclusions, and their decision making that was based upon the conclusions they reached. In the following summaries of the subjectivity of each of these scientists, we shall speculate in some cases as to how they might have used Bayesian analysis. In particular, we shall speculate on how they might have used such analysis to compare competing models of the phenomena under investigation.

Aristotle (384-322 B.C.) was a dedicated and careful observer of the natural world. Nevertheless, his belief in final causes and other overarching theories sometime led him to over-generalize or otherwise distort his observations. For example, he wrote that human beings are superior to other animals in purity of blood and softness of flesh. Because he believed that males are better equipped than females with offensive and defensive weapons, he concluded that worker bees are male (when in fact they are female). He over-generalized his observation that extremely light objects fall more slowly than do heavy ones to conclude that the speed of falling is generally related to the weight of an object. (We now understand that the difference in speed between extremely light and heavy objects only occurs because the effect of air resistance is more obvious for very light objects.) He described detailed observations that purport to show, mistakenly, that the heart is the first part of the embryo to develop. These mistaken observations lent support to his doctrine that the heart is the principle of life, the seat of the soul, of locomotion, and of what we now call higher mental functions. Since Aristotle was more inclined to speculate about scientific facts than to collect data to support his conclusions, we shall not conjecture in his case about how he might have handled probability.

Galileo (1564-1642) seems to have been convinced a priori of the law of motion that the distance fallen by a dropped object is proportional to the square of the time elapsed from dropping well before he conducted experiments to demonstrate the law. Indeed, he seems to have derived the law from a set of false premises. Further, there is some doubt as to whether the instrumentation of his day would have permitted Galileo to carry out the experiments he claims to have done, or at least to have achieved results of sufficient accuracy to prove his thesis, had he not already been convinced of its truth. Despite his position that seemed to challenge the Church's insistence on a geocentric universe, Galileo's own belief in cosmic order prevented him from accepting Kepler's substitution of elliptical planetary orbits for the more cosmically perfect circular ones. This rejection barred Galileo from full formulation of the inertial law, eventually formulated by Newton.

Galileo was forced to entertain two opposing theories of how Earth travels through the universe: does Earth travel around the Sun, or does the Sun travel around the Earth?

Copernicus and Kepler had already concluded that the heliocentric theory was to be preferred to Ptolomy's geocentric theory. Galileo's professional career was dominated by his life-threatening confrontation with the Catholic Church over which of these two theories should be accepted as truth. Putting political and religious considerations aside, Galileo might have calculated the (posterior) odds favoring each theory from a Bayesian viewpoint by comparing the (subjective) probabilities of each

[1] Adapted from Section 4.14 of, "The Subjectivity of Scientists and the Bayesian Approach", by S. James Press and Judith M. Tanur (2001), New York: John Wiley and Sons, Inc.

of the two theories to see how they compared. The powerful political position of the Church at that time, and its strong and long-held fixed belief in geocentrism, suggest that even if such a calculation had yielded results that showed the probability of the heliocentric theory overwhelming the probability of the geocentric theory, it still would have been difficult to convince the Church authorities. William Harvey (1578-1657) used what he called a "meditation" on the amount of blood issuing from the heart over a period of time to deduce that such a large quantity could not be manufactured or stored in the body; hence he was led to the notion of circulation. Having convinced himself of the reality of circulation and thus the necessity for blood to flow from arteries to veins, he was willing to believe (correctly) in the existence of a connecting network of capillaries without ever having physically observed such a network. In his work on the reproductive mechanisms of animals, Harvey was unable to identify a human ovum, but was sufficiently confident of the generality of his findings in lower animals that he subjectively (and correctly) generalized them to the entire animal kingdom.

Harvey confronted Galen's earlier theory about the heart, aorta, and all the arteries, that the blood pulsates through the body in an ebb-and-flow pattern, and certainly doesn't travel in a closed, one-directional path, by examining the implications of such a theory. He actually made calculations about what such a pulsation theory would imply about the buildup of blood in various places in the body. And he collected data that would bear on both the pulsation and circulation hypotheses. Had Bayesian methods of analysis been available he might have formed the posterior odds ratio for the probability of the "pulsation model" for blood compared with the probability of the "circulation" model he was considering. A ratio much less than 1 would have argued very strongly for the advocacy of the circulation model, which might have been very convincing to Harvey's colleagues. Harvey might also have applied such a probabilistic approach to comparing his belief in the old Aristotelian theory of "epigenesis" (the formation of a fetus by the addition of one part after another) with the more commonly-held thinking of his time, the theory of "pre-formation" (all parts of the animal fetus were present, if invisible, from the start). His data, obtained from numerous dissections, strongly supported his belief.

Subjectivity seems to have entered the work of Sir Isaac Newton (1642-1727) at a later point in the scientific process. Between the first and second editions of the *Principia*, Newton has been found to have made several changes to make the data he reported seem to fit more closely to the theories he was advocating. He reported a calculated value of the speed of sound that agreed too exactly with a value reported by another investigator, despite the fact that the earlier value was the average of many distinct measurements. Newton claimed that water vapor constitutes ten percent of air (in fact, it does not, but varies with temperature, pressure, and geographic location). He further claimed, without empirical support, that sound is not propagated through water vapor. To compensate, Newton arbitrarily increased the calculated speed of sound by 10%. Numbers purported to be data on the precession of the equinoxes were altered to better fit some corrected mathematics. Because Newton was sure of the truth of his theories, he saw his task as that of convincing his peers, and hence sometimes altered his experimental results to make them seem to support the theory more strongly. But excluding his work in alchemy and other metaphysical researches, he got his physical laws right (although they were somewhat modified later by Einstein). A major issue of model comparison concerned Newton during most of his career, the issue of the physical nature of light. Should light be thought of as corpuscular (made up of tiny particles), or as a wave motion? Newton was mostly persuaded during his early years that light should be thought of as a wave motion (and he studied the interference patterns now called "Newton's rings" which involve wave-motion thinking). Later in his career he switched positions and then thought about light as made up of particles whose behavior could be predicted by his theory of the laws of motion of material bodies. In this matter he was to be pitted against Christian Huygens, another famous scientist of the time, who strongly believed in the wave theory of light. The issue remains an arguable matter today. In what is called the dual nature of light (a theory partially attributable to Max Planck), light sometimes seems to act one way, and sometimes the other. Would a Bayesian model comparison have helped?

We believe it might have, and might still, but such a probability comparison does not yet seem to have been carried out.

For Antoine Lavoisier (1743-1794), there is a sense in which his entire corpus of work, consisting as it did primarily of a theoretical synthesis of the experimental work of others, constitutes an exercise in subjectivity. He spent a good part of his career refuting the phlogiston theory (the mistaken theory that all flammable substances contain something called phlogiston that is given off when they burn). Indeed, he was the originator of the antiphlogiston theory of combustion. Yet he was not above invoking explanations drawn from phlogiston when these seemed convenient, and he advocated the imponderable "caloric" to help explain the varying states of matter. Not only may Lavoisier have plagiarized the experiments of others, claiming to have repeated those experiments but perhaps not having actually done so, but experiments that he reported having done were sometimes carried out after the publication in which they were cited. Further, the accuracy of experimental results was sometimes exaggerated, and the number of replications achieved similarly inflated. Like Newton, Lavoisier was subjectively convinced of the accuracy of the scientific system he was proposing, the systematization of chemistry, and sometimes gave in to the temptation to embellish the empirical support for that system. But the system and most of its details, especially the understanding of combustion and other forms of oxidation, were sufficiently correct to earn Lavoisier the sobriquet, "father of modern chemistry."

The argument between the antiphlogistians and the advocates of the phlogiston theory of combustion might have been resolved more readily had the posterior probability calculations been presented to researchers caught up in Lavoisier's chemical revolution. Again, the issue was the selection of the correct model to use in order to come closer to truth.

Alexander von Humboldt's (1769-1859) subjectivity influenced him, early in his career, to interpret geological observations as evidence supporting the Neptunist theory of the origin of rocks in sedimentation. Later, more extensive observations and a change in his theoretical orientation led to a reversal of the interpretation of the same observations as evidence in favor of the volcanic origin of the rocks (the Plutonist theory). Early in his scientific career, von Humboldt set out to demonstrate the harmoniousness of nature; accordingly, his voluminous writings repeatedly make the point that nature is one great whole, in which plants, animals, and geological and meteorological phenomena as well as human beings and their culture fit coherently. Bayesian statistical methods of analysis, had they been available at the time, could readily have compared the Neptunist and Plutonist theories.

Michael Faraday (1791-1867), like von Humboldt, sought a unity in nature. Under the influence of his religion of Sandemanianism, Faraday believed in the simplicity and integration of nature. This led him to attempt to establish the unity of the forces of magnetism, electricity, and gravitation. He achieved the conversion of electrical to mechanical energy and established the equivalence of all types of electricity. His conviction led to his persistence through many years of attempts before he was successful in showing the conversion of magnetism to electricity. He continued to believe in the relationship between electricity and gravity, although he was unable to establish that relationship. Faraday even left a letter to be opened 60 years after his death that carried the idea of unity of forces further with speculation on the existence of electromagnetic waves analogous to those of the ocean.

Charles Darwin (1809-1882) believed theory must guide observations despite his professed belief in Baconian induction. His synthesis of his observations led him to a theory of evolution powered by a mechanism of natural selection, which would yield gradual changes in organisms. This subjective belief led him to explain gaps in the fossil record where intermediate forms should have been represented as being merely temporary gaps in data gathering, to be filled as paleontologists explored further. It also led him to disbelieve then current calculations of the age of Earth, since the accumulation of gradual changes would require more time than those calculations provided. Scientific opposition to Darwin's proposed mechanism of natural selection was coupled with the mistaken belief (which Darwin shared) that the material of heredity was "infinitely divisible." If that were the case, the substance in the human body that passed heredity characteristics, such as eye color, on to the next generation could pass on any degree of "blueness," for example, as opposed to just "blue" or "not blue." An implication of this mistaken belief was that a change (mutation) in an organism would be diluted through interbreeding. That implication caused Darwin to change his mind about natural selection. His later writings gave but scant importance to natural selection and stressed instead a purely subjective notion of pangenesis, which implied the inheritance of acquired characteristics. Neither Darwin nor any of his contemporaries or successors could find any empirical evidence for the inheritance of acquired characteristics, called the Lamarckian view. The lack of empirical support for Lamarckian mechanisms was coupled with the re-discovery of the work of Gregor Mendel, which showed that the material of heredity is "particulate," that is, not infinitely divisible but discrete and thus that mutations can be stored in the genes without being diluted. These findings eventually caused the scientific community to revive natural selection as the mechanism for evolution, despite the fact that Darwin had abandoned the idea.

These two models of inheritance, "pangenesis," implying the inheritance of acquired characteristics (the Lamarckian view), and "natural selection," were opposed to one another. There were considerable data available that bore on the two models. They could easily have been compared probabilistically, as part of the scientific analysis. Such an analysis, and related analyses about the model of evolution versus. the model of biblical genesis have not yet been carried out, and the arguments surrounding such competing models still rage. As in the case of Galileo and the heliocentric and geocentric theories, the arguments regarding natural selection and evolution go far beyond science and mathematics by pitting religious belief against the opinions of the scientific community. While it is likely that probability would have helped somewhat in both Galileo's time, and in Darwin's, the issue of convincing the public is one that pits science against religion.

Louis Pasteur (1822-1895) has been shown within recent years to have permitted his strong prior beliefs influence his clinical treatments and his reports of data he falsely claimed to have acquired through experimentation. On the basis of theory and its demonstrated effectiveness in the case of the microbe for chicken cholera, Pasteur believed that atmospheric oxygen would be effective in attenuating the virulence of other microbes. He was extremely enthusiastic about this theory of the usefulness of atmospheric oxygen. Hence in reporting a highly successful public test of the effectiveness of a vaccine against anthrax, Pasteur claimed that he had prepared the vaccine using atmospheric oxygen, when in fact it had been produced using potassium bichromate. Similarly, when he believed he had hit upon a process that would produce immunity to rabies, Pasteur felt himself justified in using that vaccine on two young boys who had been bitten by a rabid dog. He claimed that he had already successfully immunized 50 dogs with this same vaccine; his recently revealed laboratory notebooks make it clear that he had barely begun trials of this vaccine and had as yet no results at all. In his work to disprove the doctrine of the spontaneous generation of life, Pasteur is seen to have suppressed the results of experiments that had what he considered "errors," that is, data that might lend support to the theory he was trying to disprove. We now know that Pasteur's experimental procedure generated insufficient heat to kill all the microorganisms originally present in his experimental material, so that the "erroneous" results do not in fact support the idea of spontaneous generation, but Pasteur himself did not have this explanation available. Regardless, Pasteur's sterilization process was eventually justified, and resulted in the pasteurization process that has currency today.

One of the several model comparisons that Pasteur was involved with was the one that examined the theory of "spontaneous generation" and he confronted it with the theory that life can only derive from other life. He introduced the notion of sterilization (now called pasteurization) to prove his point. Would probabilistic analysis have helped his cause? Probably, but despite all of his brilliance and flashes of scientific insight, Pasteur's approach to data was to suppress them if they didn't agree with his pre-conceptions of what they ought to be. More objective Bayesian analyses by others of data analogous to Pasteur's would surely have helped Pasteur make his case more convincing.

Partly because Sigmund Freud (1856-1939) used subjectivity so pervasively in his work, some observers believe that his work is not science at all. He used data based on patients' free associations and recounting of dreams to interpret their subconscious thoughts, and thus derive what he considered universal laws indicating that the sexual urge and the libido are the driving forces behind all repression and inner conflict. Observers of Freud's work have stressed that his ideas evolved in a mind suffused with a sense of personal destiny and that maintained a lively interest in the occult. His attention to mysticism perhaps stemmed from his father's Chassidic mysticism and from his own heavy use of cocaine. Because they argue that Freud never scientifically tested his claims for the causal links he proposed between sexual repression and neurosis, and between the processes of psychoanalysis and cure, some critics feel Freud's work is pure subjectivity (rather than the required combination of subjectivity and objectivity found in all good science).

Freud's data are questionable in a scientific sense since he did not carry out controlled experiments and his observations were totally subjective assessments of what was happening to his patients. But that has been the nature of most of the field of psychoanalysis, although there are positive signs that the stance may be undergoing change in the appropriate direction.

Marie Curie (1967-1934) was long sustained in her investigation of radium by her belief that the new substance was an element. But her belief that elements could not be transmuted to other elements (even through radioactive decay) kept her from being able to explain the origin of radioactivity, although she (together with her husband, Pierre) originated the insight that radiation was an atomic property. The fact that the elements with which she was working (not only radium, but polonium as well) had long half lives made it extremely difficult for Curie to measure the loss of weight and energy due to radioactive decay. Thus, despite her toying in a 1900 publication with the idea of transmutation, Marie Curie's strong subjective prior belief in the established principles of physics, coupled with her strong belief in what her data told her, kept the next insight from her. A Bayesian model comparison of alternative theories purporting to explain the radioactivity phenomenon would probably have suggested new and exciting paths of discovery to Madam Curie.

Albert Einstein (1879-1955) was a *theoretical* physicist in the sense that he derived his beliefs about the physical world not only from a profound understanding of the way the physics of the world works, but also from the mathematical models that he developed of relationships that physical phenomena should obey. His equations governing the special and general theories of relativity, for example, were used to make corrections to Isaac Newton's laws of motion, and to generate predictions about what might be observed in appropriate hypothetical circumstances, as when traveling at speeds near the speed of light. But he never experimentally tested any of his theories; he merely stated them subjectively and mathematically, fully expecting them to correctly characterize what would be observed in the real world. He also believed that if data were collected to test his theories, and if the data contradicted theory, he would be inclined to reject the data as erroneous, because they contradicted the theory that he simply knew a priori to be true.

Einstein was fully capable of carrying out any mathematical analysis of his choosing. So to examine competing theories of physics using probability and posterior odds ratios would have been well within his grasp. He even used probability to develop the (predictive) law of Brownian motion, one of his famous papers of 1905. But he had difficulties with collecting his own data. He was not an experimentalist. So it was not his approach to decide about any physical laws on the basis of being convinced by a preponderance of scientific data–expressed probabilistically or otherwise. Einstein just intuitively and mathematically decided how things ought to be, and then enunciated what the general principles had to be. Probability considerations are very unlikely to have been much of a cogent force on his own beliefs, although they might have strengthened the scientific beliefs of others in his theories (such as in the theory of general relativity) at the earlier stages in his career.

We can discern several themes in this recital of the earlier uses of informal subjectivity in science, and the use of formal probability to examine competing scientific hypotheses.

First, we see generalizations from data (induction), precisely as Bacon had prescribed (and contrary to Popper's views involving falsification). But sometimes, these generalizations go well beyond the data at hand,

and can lead either to correct inferential leaps or mistaken over-generalizations. We have seen Harvey correctly assume that the origin of all life is in the ovum, even though he was unable to understand the significance of the ovaries in mammals. But we have also seen Aristotle over-generalize from watching extremely light and heavy objects fall. He concluded that heavy objects tend to fall toward the earth and light ones tend to rise and thus that the speed of fall is related to an object's weight.

Second, we have seen instances in which a scientist, overwhelmed by a theory, is apt to see what the theory prescribes (or is unlikely to see what the theory denies). Marie Curie, guided by a belief in the conservation of energy, failed to identify the mechanism that gives rise to radioactivity. And Albert Einstein insisted that any data that might seem to contradict his theory would prime facie be mistaken. Note how this statement echoes Galileo's expression of awe at learning that the adherents of the Copernican (heliocentric) system were able to ignore the evidence of their own senses (that the Sun appears to rotate around Earth) to conclude that Earth revolves around the Sun.

Third, we have seen instances in which a scientist becomes sufficiently convinced of the correctness of the line of investigation being followed that a single-minded stubbornness develops. We have seen Marie Curie insisting that her newly discovered substance, radium, is an element, and we see Michael Faraday laboring for a decade to demonstrate the induction of electricity from magnetism.

Finally there are instances of subjectivity that fall on the border of outright fraud - and instances that fall squarely within the domain of the fraudulent. To range outside of the 12 scientists we have been discussing for a moment, Johannes Kepler, Gregor Mendel, Robert Millikan and Cyril Burt reported data that were "too good to be true", to extend use the term employed by Fisher (see Fisher, R. A., 1936, "Has Mendel's Work Been Rediscovered?", Annals of Science, 1, 115-137) to describe his view of the work of Mendel, but that in each case fit the theory that the scientist was trying to prove or to demonstrate. The data doctoring in three of these cases was in the service of a theory now believed to be true; in Burt's case the jury is still out. Sir Isaac Newton altered data between editions of his masterpiece, *The Principia*, in order that his theoretical contentions appear in the best possible light. Both Antoine Lavoisier and Louis Pasteur claimed to have carried out experiments that were either incomplete or even not yet begun at the time they published the supposed results. We do not condone their bending, or even breaking, the accepted rules of honesty applicable no less in the practice of science than more broadly in normal human interaction. Nevertheless, we must note that Newton, Lavoisier, and Pasteur were most often correct in their scientific conclusions, and that they enormously advanced the cause of science with their thinking.

Subjectivity occurs, and should occur, in the work of scientists; it is not just a factor that plays a minor role that we need to ignore as a flaw that sometimes creeps into otherwise "objective" scientific analysis. Total objectivity in science is a myth. Good science inevitably involves a mixture of subjective and objective parts.

MORE ABOUT THOMAS BAYES

by Anthony O'Hagan
a.ohagan@sheffield.ac.uk

Since writing in the last Bulletin about my very amateur piece of historical investigation on Thomas Bayes and Richard Price, I have heard about two much more extensive and authoritative studies by colleagues in Canada and South Africa. I am sure that many ISBA members would like to know about these.

David Bellhouse has discovered some manuscripts that throw light on Bayes's status in contemporary mathematical circles, and has written a fascinating biography of Bayes. This and a technical article on the manuscripts themselves can be found on David's website at `www.stats.uwo.ca/faculty/bellhouse/bayespage.htm`.

Then you can look forward to a substantial book on Bayes written by Andrew Dale (`DALE@nu.ac.za`) and due to be published by Springer-Verlag next year. Andrew says, "It should run to about 600 pages, I think. There will be some chapters on the Bayes family (as much as I have been able to find) and on Thomas's time in Tunbridge Wells. Then chapters on the works, and finally something about Bunhill Fields, the Bayes family vault and wills of some of the Bayes family".